

Harith Al-Safi

Machine Learning Engineer

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📄 About me

A Software Engineer with 5 years of specializing in Machine Learning, React.js, Java Spring Boot and Cloud Technologies. Proven record in leading technical teams, and driving cost optimizations across **startup** to **enterprise**.

📁 Experience

Full Stack Engineer

📅 Sep 2024 - Ongoing (16+ Months)

BT Group

📍 Manchester, UK

- Delivered 5 **Kafka-based Spring Boot microservices** with MongoDB/PostgreSQL, Netflix Conductor, and Pulumi
- Mentored and examined 5+ apprentices on agile development with **Java 21** Spring Boot and **AWS Lambda**
- Achieved **30% AWS cost reduction** by automating infrastructure with TypeScript and optimizing ECS cluster
- Re-architecture Spring Boot APIs with JWT, Azure AD, and **role-based access** achieving **enterprise grade security**
- Led **React.js to Next.js** migration of 5K+ lines, adding Cypress testing and OpenAPI to TypeScript **code gen**
- Managed regional Graduate Network organizing and coordinating monthly tech talks for **30+ attendees**

Robotics Engineer

📅 Dec 2023 - Mar 2024 (4 Months)

Robot Fight League

📍 Leeds, UK

- Implemented a **control system** using Arduino and **C++** for the robot's movement, sensors and pneumatic system
- Tested obstacle detection and tracking using **OpenCV** and Python for the robot's **autonomous navigation** system
- Led **cross-functional** team of 5 engineers to 1st place finish, coordinating **hardware-software** integration

Software Engineer (Placement Year)

📅 July 2022 - July 2023 (12 Months)

Johnson Controls

📍 London, UK

- Invented a **scripting language with a compiler** for fire detection, increasing development productivity by 40%
- Managed a team of 10 interns to create a **business plan** and ended up placing 2nd amongst the whole company
- Administered C/C++ projects while fixing bugs, adding features, refactoring and **migrating 90% of codebase to C#**
- Developed C# .NET libraries for fire detection simulations and data analysis improving operations by 32%
- Collaborated on an **analytics web dashboard with React.js** for predictive maintenance with dev ops and CI/CD
- Maintained Python automation scripts for automated builds of .NET applications and processing data with Azure

Machine Learning Engineer

📅 Dec 2021 - May 2022 (6 Months)

Paperound - Freelance

📍 Leeds, UK

- Developed **signal processing filters** and modulators using SciPy Python and Fourier analysis for RF signals
- Built and trained a **reinforcement neural network** using TensorFlow, KerasRL for predicting market movements
- Applied **data visualization** using Python Matplotlib to present analytical insights to non-technical stakeholders
- Built production-ready **LLM & RAG** inference APIs with Docker and Flask for low-latency model serving.
- Developed parallel processing system using **C++ OpenMP** to **accelerate polynomial regression** computations

Information Technology Administrator

📅 Jun 2020 - Jun 2021 (12 Months)

Mesopotamia Group

📍 Amman, Jordan

- Architected **scalable file server infrastructure** using Apache, Nextcloud, and Ubuntu in production environment
- Automated system administration workflows using **Python, Bash scripting**, and network security protocols
- Developed **high-availability load balancer with C/MPI** achieving 34% improved system uptime
- Built **predictive analytics pipeline using Python** for proactive cybersecurity threat detection
- Designed cloud networking solutions implementing TCP/IP, routing protocols, and infrastructure optimization

📖 Education

BEng (Honours) in Electronics and Computer

📅 Sept 2019 - July 2024

University of Leeds (UoL)

✅ First-Class Honours (4.0/4.0 GPA)

- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Data Science
- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design

Skills and Languages

Technical:

PythonC \ C++C#TypeScriptJavaBashMatplotlibTensorFlowScikit-learnNumpySciPyPandasNLPReact.jsNext.jsExpress.jsReduxTailwind CSSStripeAuth0ClerkSpring BootKafka.NETDockerJiraGitMongoDBPostgreSQLRedisSQLiteJenkinsFirebase FirestoreAWS (RDS, Lambda, ECS, EC2, S3)Azure (AD, MSAL, Serverless)Pulumi

Soft:

ManagementLeadershipMotivatedDeterministicCommunicationProblem solver

Languages: Arabic (Full Professional)English (Full Professional)Spanish (Elementary)

Publications

Vega: LLM-driven Intelligent Chatbot Platform for Internet of Things Control and Development 2025

MPDI Sensors - 10.3390/s25123809 - 1st Author

- Developed a full stack **Chatbot** web application with TypeScript, React.js, Next.js, **TailwindCSS** and **Reddit**
- Programmed language processing algorithms using **multi-modal large language models** to execute user messages
- Implemented a **Python Flask** REST Server with **SQLite** on the **Raspberry Pi** to communicate with the chatbot
- Engineered a **circuit** with sensors, camera, fans, LCD and buttons to execute user's commands from the chatbot

Projects

Green ML: Machine Learning for Green Bond Syndication May 2024 - Aug 2024 Personal

- Engineered bond syndication **prediction model** using **XGBoost** and network analysis on 3K+ institutional holders
- Implemented **parallel processing** with Ray to extract network features from adjacency matrices with **70% speedup**
- Built machine learning pipeline achieving **0.95+ AUC-ROC** for predicting bond syndication collaboration patterns
- Processed **Bloomberg financial data** using **pandas**, **NetworkX** for multi-regional investment analysis

CV Studio Oct 2024 - Ongoing Personal - Beta on cvstudio.ai

- Built **full-stack** resume-improvement web app, powered by large language model, serving **hundreds of active users**.
- Implemented **live resume editing** with ShadcnUI and Typst **WebAssembly** PDF compilation for real-time render.
- Integrated **Clerk authentication** and **Stripe subscription** payments for secure user access and monetization
- Designed **text embedding and reranking** using HuggingFace to score resumes against job descriptions

MCPP: High Performance C++ Library for Numerical Computation Apr 2021 - Oct 2022 Personal

- Created a **cross-platform math library** for C++ used for matrix algebra, statistics and machine learning.
- Engineered with **CI/CD** using **Jenkins**, unit tests, Docker build, **Cmake** and Doxygen documentation
- Improved performance by 40% using **parallel computing** with **OpenMP**, **OpenCL** and **CUDA**

Professional Courses

Data Analysis with Python Oct 2021

- Develop Python code for cleaning and preparing data for analysis
- Perform exploratory **data analysis** and analytical techniques to datasets using Pandas, Numpy and Scipy
- Manipulate data using dataframes, understand data distribution, perform correlation and create data pipelines
- Build and evaluate **regression models** using machine learning scikit-learn for prediction and decision making

Databases and SQL for Data Science with Python Sept 2021

- Analyze data within a relational database on the cloud using **IBM DB2**, SQL, Python
- Construct **SQL CRUD** operations and compose views, transactions, stored procedures and joins.

Python for Data Science, AI & Development Aug 2021

- Demonstrate proficiency in using Python libraries such as Pandas, **Numpy**, and Beautiful Soup
- Access web data using APIs and **web scraping** from Python in Jupyter Notebooks.